Unit 05: Introduction to Mining Tools

CONTENTS

Objectives

After this lecture, you will be able to

- Understand the basics of rapidminer and its products.
- Learn the working of the Weka tool.
- understand various features of a rapid miner and Weka Tool.
- Know the Interface of RapidMiner and Weka.

Introduction

Several open-source and proprietary-based data mining and data visualization tools exist which are used for information extraction from large data repositories and for data analysis. Some of the data mining tools which exist in the market are Weka, Rapid Miner, Orange, R, KNIME, ELKI, GNU Octave, Apache Mahout, SCaViS, Natural Language Toolkit, Tableau, etc.

5.1 RapidMiner

RapidMiner is an integrated enterprise artificial intelligence framework that offers AI solutions to positively impact businesses. RapidMiner is widely used in many business and commercial applications as well as in various other fields such as research, training, education, rapid prototyping, and application developmentAll major machine learning processes such as data preparation, model validation, results in visualization, and optimization can be carried out by using RapidMiner.

Rapidminer comes with :

- **Over 125 mining algorithms**
- Over 100 data cleaning and preparation functions.
- Over 30 charts for data visualization,and selection of metrics to evaluate model performance.

5.2 Installing Rapidminer on your machine

- The latest version of Rapidminer Studio is V7, it can be downloaded from https://rapidminer.com/products/comparison/
- For Windows: download the rapidminer-install.exe and install.
- Defaults install it to C:\program files and add it to the start>programs menu.
- For mac: download the .dmg and add it to your applications folder.

RapidMiner Products

There are many products of RapidMiner that are used to perform multiple operations. Some of the products are:

- **RapidMiner Studio:**With RapidMiner Studio, one can access, load, and analyze both traditional structured data and unstructured data like text, images, and media. It can also extract information from these types of data and transform unstructured data into structured.
- **RapidMiner Auto Model:**Auto Model is an advanced version of RapidMiner Studio that increments the process of building and validating data models. You can customize the processes and can put them in production based on your needs. Majorly three kinds of problems can be resolved with Auto Model namely prediction, clustering, and outliers.
- **RapidMiner Turbo Prep:**Data preparation is time-consuming and RapidMiner Turbo Prep is designed to make the preparation of data much easier. It provides a user interface where your data is always visible front and center, where you can make changes step-by-step and instantly see the results, with a wide range of supporting functions to prepare the data for model-building or presentation.

TOOL CHARACTERISTICS

- **Usability: E***asy to use*
- **Tool orientation:**The **t**ool is designed for general-purpose analysis
- **Data mining type:**This tool is made for *Structured data mining, Text mining, Image mining, Audio mining, Video mining, Data gathering, Social network analysis*.
- **Manipulation type:**This tool is designed for *Data extraction, Data transformation, Data analysis, Data visualization, Data conversion, Data cleaning*

5.3 Features of RapidMiner

- **Application & Interface:**RapidMiner Studio is a visual data science workflow designer accelerating the prototyping & validation of models.
- **Data Access**: With RapidMiner Studio, you can access, load, and analyze any type of data – both traditional structured data and unstructured data like text, images, and media. It can also extract information from these types of data and transform unstructured data into structured.
- **Data Exploration:** Immediately understand and create a plan to prepare the data automatically extract statistics and key information.
- **Data Prep:**The richness of the data preparation capabilities in RapidMiner Studio can handle any real-life data transformation challenges, so you can format and create the optimal data set for predictive analytics. RapidMiner Studio can blend structured with unstructured data and then leverage all the data for predictive analysis. Any data preparation process can be saved for reuse.
- **Modeling**: RapidMiner Studio comes equipped with an un-paralleled set of modeling capabilities and machine learning algorithms for supervised and unsupervised learning. They are flexible, robust and allow you to focus on building the best possible models for any use case.

- **Validation:**RapidMiner Studio provides the means to accurately and appropriately estimate model performance. Where other tools tend to too closely tie modeling and model validation, RapidMiner Studio follows a stringent modular approach which prevents information used in pre-processing steps from leaking from model training into the application of the model. This unique approach is the only guarantee that no overfitting is introduced and no overestimation of prediction performances can occur.
- **Scoring:**RapidMiner Studio makes the application of models easy, whether you are scoring them in the RapidMiner platform or using the resulting models in other applications.
- **Code Control & Management:**Unlike many other predictive analytics tools, RapidMiner Studio covers even the trickiest data science use cases without the need to program. Beyond all the great functionality for preparing data and building models, RapidMiner Studio has a set of utility-like process control operations that lets you build processes that behave like a program to repeat and loop over tasks, branch flows and call on system resources. RapidMiner Studio also supports a variety of scripting languages.

5.4 How does it work?

You visually design a data mining process. A process is like a flow chart for mining operators.

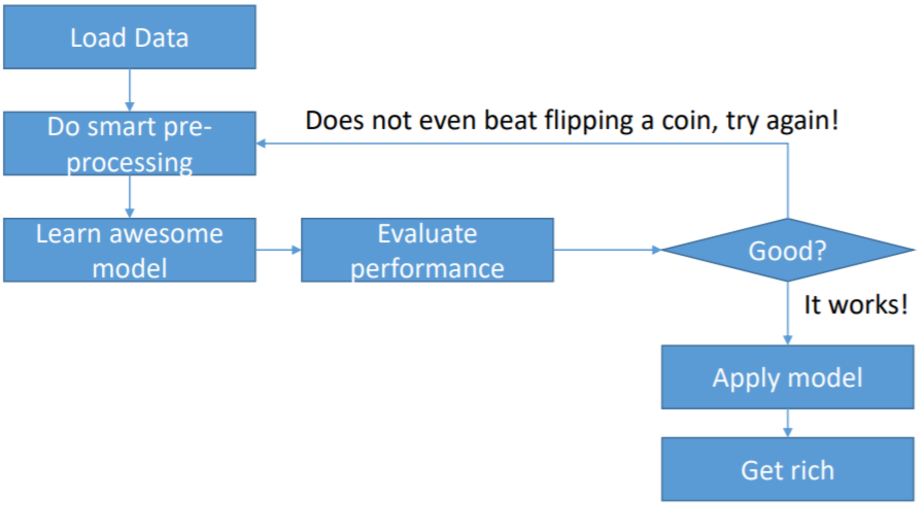

Figure 1: Working of RapidMiner

All data that you load will be contained in an example set. Each example is described by Attributes (a.k.a. features) and all attributes have Value Types and their specific Roles. The Value types define how data is treated

- Numeric data has an order (2 is closer to 1 than to 5)
- Nominal data has no order (red is as different from green as from blue)

Table 1: Different Value Type

Value Type	Description
binominal	Only two different values are permitted
polynomial	More than two different values are permitted
integer	Whole numbers, positive and negative
real	Real numbers, positive and negative
date_time	Date as well as time date Only date-time Only time

Roles define how the attribute is treated by the Operators.

Table 2: Types of Roles

Role	Description
Id	A unique identifier, no two examples in an example set can have the same value
Regular (default)	Regular attribute that contains data
Label	The target attribute for classification tasks
Weight	The weight of the Examples concerning the label
Cluster	Created by RapidMiner as the result of a clustering task
Prediction	Created by RapidMiner as the result of a classification task

The Repository

This is where you store your data and processes.Only if you load data from the repository, RapidMiner can show you which attributes exist. Add data via the "Add Data" button or the "Store" operator. You can load data via drag 'n' drop or the "Retrieve" operator. If you have a question starting with "Why does RapidMiner not show me …?" Then the answer most likely is "Because you did not load your data into the Repository!"

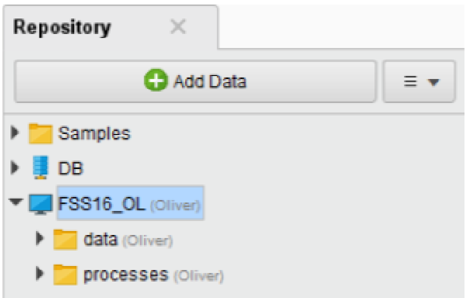

Figure2:Reository

A repository is simply a folder that holds all of your RapidMiner data sets (we call them "ExampleSets), processes, and other file objects that you will create using RapidMiner Studio. This folder can be stored locally on your computer, or on a RapidMiner Server.

Figure3: Store Operator

This operator stores an IO Object at a location in the data repository. The location of the object to be stored is specified through the *repository entry* parameter. The stored object can be used by other processes by using the Retrieve operator.

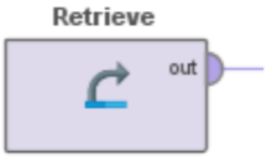

Figure 4: Retrieve Operator

Example: Themost simple process which can be created with RapidMiner: It just retrieves data from a repository and delivers it as a result to the user to allow for inspection.

Benefits of RapidMiner

The main benefits of RapidMiner are its robust features, user-friendly interface, and maximization of data usage. Learn more of its benefits below:

Robust features and user-friendly interface
RapidMiner's tools and features offer powerful capabilities for the users while at the same time are presented through a user-friendly interface that allows users to perform productively in their works from the start. Thus, each of the tools' robust components is easy to be operated. One feature of the system is the visual workflow designer which is a tool that provides a visual environment to the users. This environment is where analytics processes can be designed, created, and then deployed. Visual presentation and models can also be made and processed here. All of these can easily be done by the users because of the friendly environment.

Maximize the usage of data
The system provides the users with the right set of tools which makes relevant uses of even the most disorganized, uncluttered, and seemingly useless data. This can be accomplished by enabling the users and their team to structure data in an easy way for them to comprehend. To do this, RapidMiner offers capabilities that simplify data access and management that will empower users to load, access, and evaluate all types of data such as images and texts.

Not only does the system allows the usage of any data but it also allows them to create models and plans out of them, which can then be used as a basis for decision making and formulation of strategies. RapidMiner has data exploration features, such as descriptive statistics and graphs and visualization, which allows users to get valuable insights out of the information they gained.RapidMiner is also powerful enough to provide analytics that is based on real-life data transformation settings. This means that users can manipulate their data any way they want since they have control of its formatting and the system. Because of this, they can create the optimal data set when performing predictive analytics.

How to use RapidMiner

Use the "Design Perspective" to create your Process

- See your current Process – "Process"
- Access your data and processes – "Repository"
- Add operators to the process – "Operators"
- Configure the operators – "Parameters"
- Learn about operators – "Help"

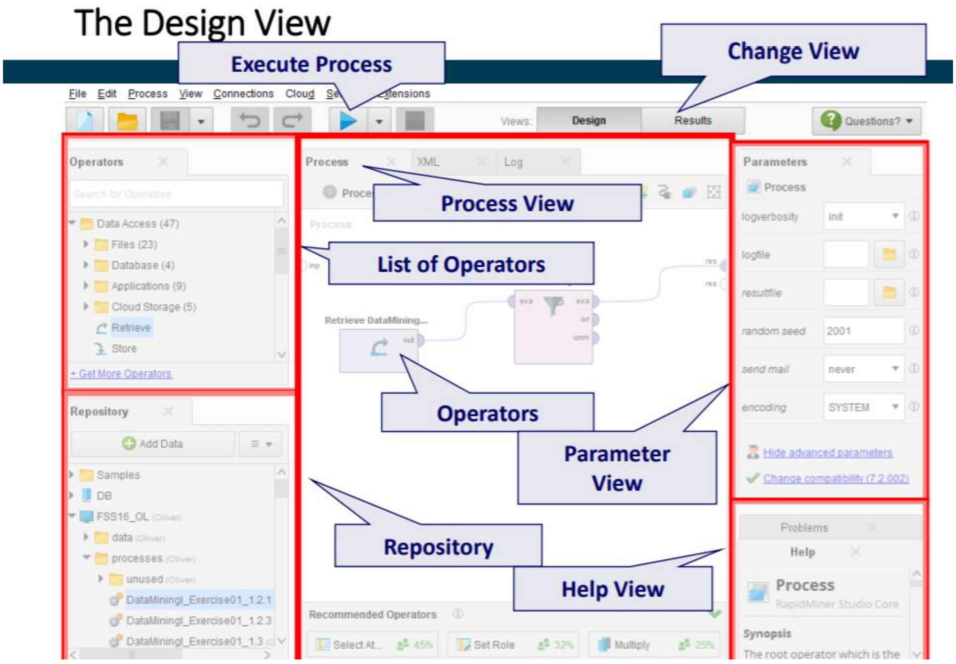

Figure 5:Design View

Use the "Results Perspective" to inspect the output

- The "Data View" shows your example set
- The "Statistics View" contains metadata and statistics
- The "Visualizations View" allows you to visualize the data

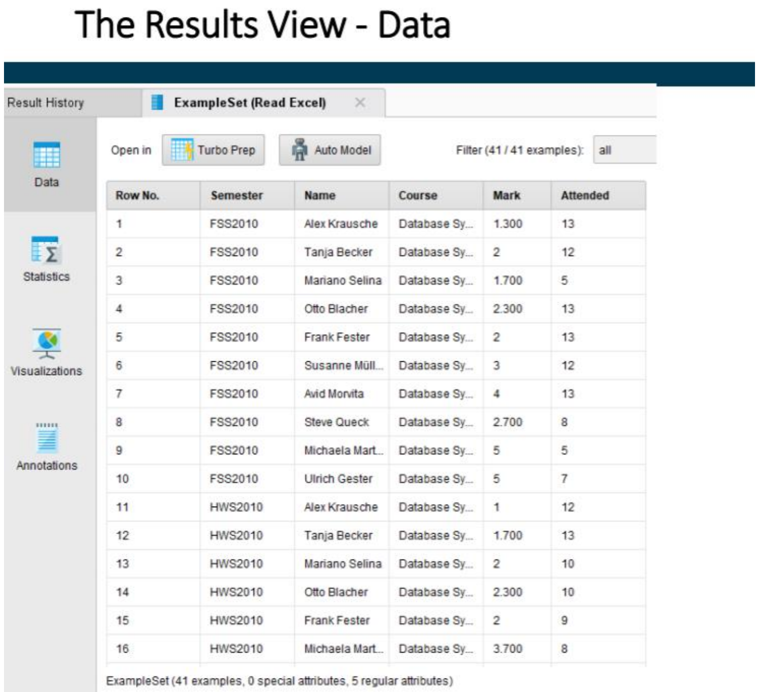

Figure 6:The Result View

Finding an operator

Once you get familiar with operator names, you can find them more easily using the filter at the top of the operator window.

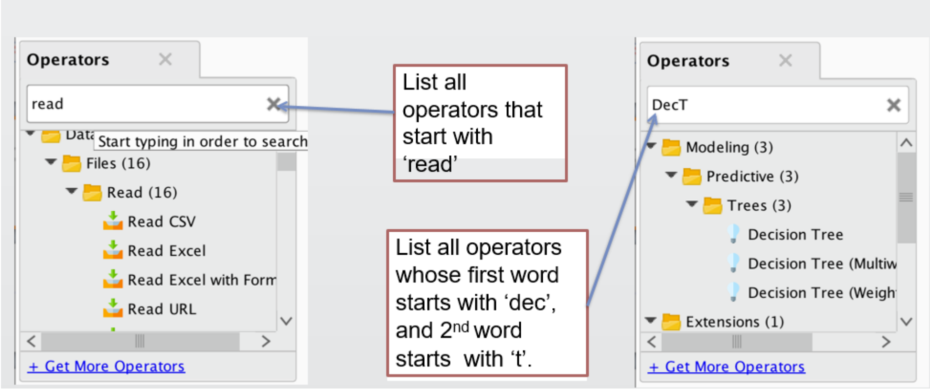

Figure 7: Ways to find an Operator

5.5 Weka

Weka is data mining software that uses a collection of machine learning algorithms. These algorithms can be applied directly to the data or called from the Java code.The algorithms can either be applied directly to a dataset or called from your own Java code. Weka contains tools for data pre-processing, classification, regression, clustering, association rules, and visualization. It is also well-suited for developing new machine learning schemes.

WEKA an open-source software that provides tools for data preprocessing, implementation of several Machine Learning algorithms, and visualization tools so that you can develop machine learning techniques and apply them to real-world data mining problems. What WEKA offers is summarized in the following diagram −

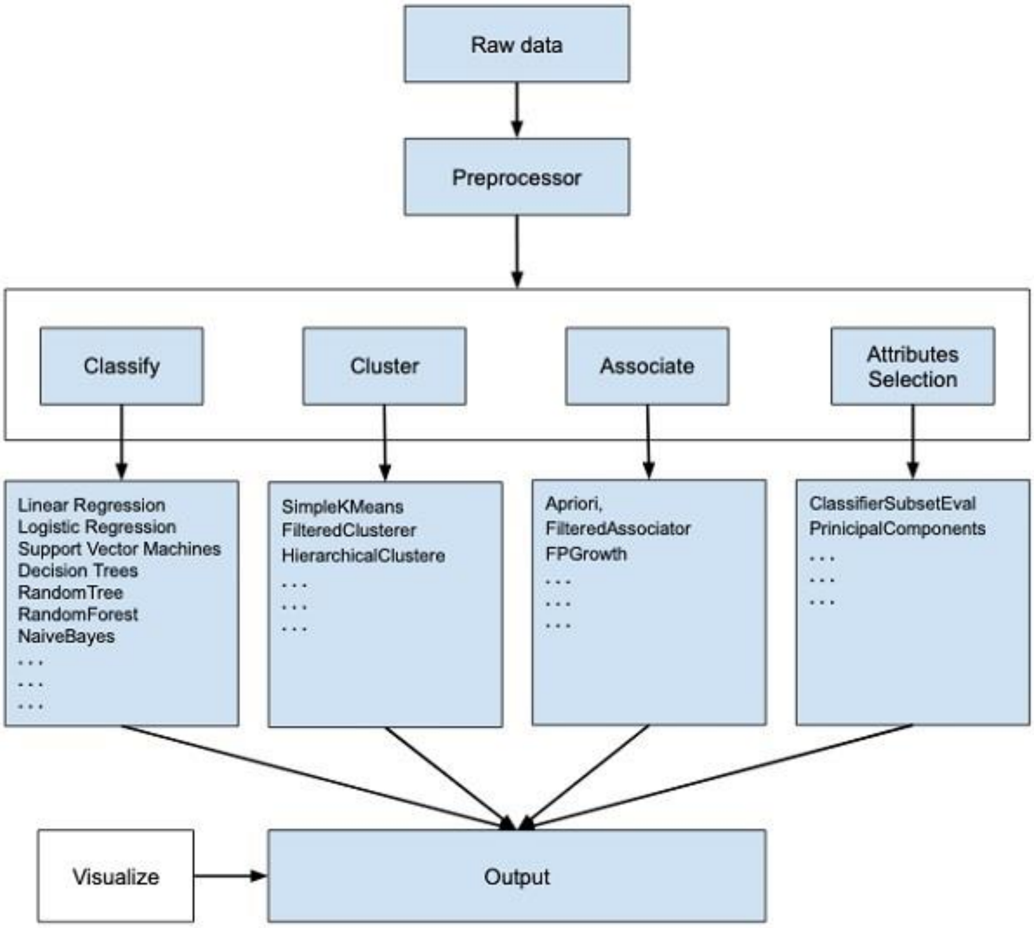

Figure 8: Working of WEKA

If you observe the beginning of the flow of the image, you will understand that there are many stages in dealing with Big Data to make it suitable for machine learning −

First, you will start with the raw data collected from the field. This data may contain several null values and irrelevant fields. You use the data preprocessing tools provided in WEKA to cleanse the data.Then, you would save the preprocessed data in your local store for applying ML algorithms.

Next, depending on the kind of ML model that you are trying to develop you would select one of the options such as Classify, Cluster, or Associate. The Attributes Selection allows the automatic selection of features to create a reduced dataset.Note that under each category, WEKA provides the implementation of several algorithms. You would select an algorithm of your choice, set the desired parameters, and run it on the dataset.

Then, WEKA would give you the statistical output of the model processing. It provides you a visualization tool to inspect the data.The various models can be applied to the same dataset. You can then compare the outputs of different models and select the best that meets your purpose.Thus, the use of WEKA results in quicker development of machine learning models on the whole.

Weka is a collection of machine learning algorithms for data mining tasks. The algorithms can either be applied directly to a dataset or called from your own Java code. Weka contains tools for data pre-processing, classification, regression, clustering, association rules, and visualization. The WEKA GUI Chooser application will start and you would see the following screen −

Figure 9:Weka GUI Chooser

The GUI Chooser application allows you to run five different types of applications as listed here −

- Explorer
- Experimenter
- KnowledgeFlow
- Workbench
- Simple CLI

When you click on the Explorer button in the Applications selector, it opens the following screen –

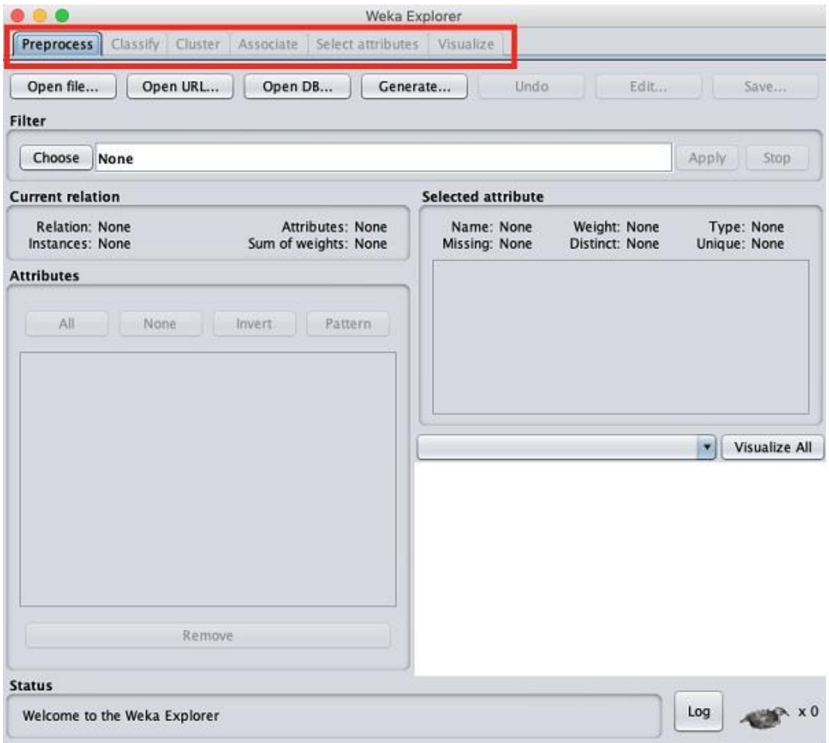

Figure 10: Weka Explorer

- On the top, you will see several tabs as listed here –
- Preprocess
- Classify
- Cluster
- Associate
- Select Attributes
- Visualize
- Under these tabs, there are several pre-implemented machine learning algorithms. Let us look into each of them in detail now.

Preprocess Tab

Initially, as you open the explorer, only the **Preprocess** tab is enabled. The first step in machine learning is to preprocess the data. Thus, in the **Preprocess** option, you will select the data file, process it, and make it fit for applying the various machine learning algorithms.

Classify Tab

The **Classify** tab provides you several machine learning algorithms for the classification of your data. To list a few, you may apply algorithms such as Linear Regression, Logistic Regression, Support Vector Machines, Decision Trees, RandomTree, RandomForest, NaiveBayes, and so on. The list is very exhaustive and provides both supervised and unsupervised machine learning algorithms.

Cluster Tab

Under the **Cluster** tab, there are several clustering algorithms provided - such as SimpleKMeans, FilteredClusterer, HierarchicalClusterer, and so on.

Associate Tab

Under the **Associate** tab, you would find Apriori, FilteredAssociator, and FPGrowth.

Select Attributes Tab

Select Attributes allows you to feature selections based on several algorithms such as ClassifierSubsetEval, PrinicipalComponents, etc.

Example:They can be used, for example, to store an identifier with each instance in a dataset.

Visualize Tab

The **Visualize** option allows you to visualize your processed data for analysis. WEKA provides several ready-to-use algorithms for testing and building your machine learning applications. To use WEKA effectively, you must have a sound knowledge of these algorithms, how they work, which one to choose under what circumstances, what to look for in their processed output, and so on. In short, you must have a solid foundation in machine learning to use WEKA effectively in building your apps.

we start with the first tab that you use to preprocess the data. This is common to all algorithms that you would apply to your data for building the model and is a common step for all subsequent operations in WEKA.

For a machine-learning algorithm to give acceptable accuracy, you must cleanse your data first. This is because the raw data collected from the field may contain null values, irrelevant columns, and so on.

First, you will learn to load the data file into the WEKA Explorer. The data can be loaded from the following sources −

- Local file system
- Web
- Database

Analyze your data with WEKA Explorer using various learning schemes and interpret received results.

We will see all three options of loading data in detail.

Loading Data from Local File System

Just under the Machine Learning tabs that you studied in the previous lesson, you would find the following three buttons −

- Open file ...
- Open URL ...
- Open DB ...

Click on the **Open file** ... button. A directory navigator window opens as shown in the following screen −

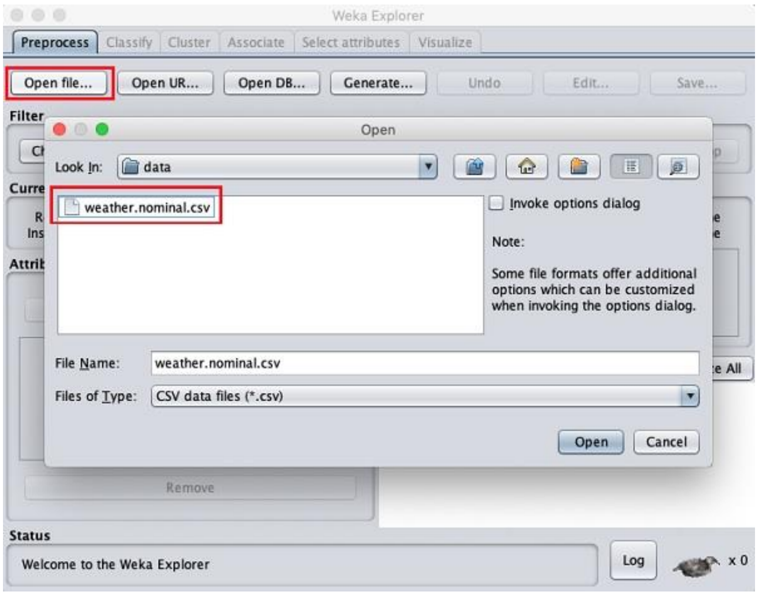

Figure 11: Opening a File using Explorer

Now, navigate to the folder where your data files are stored. WEKA installation comes up with many sample databases for you to experiment with. These are available in the **data** folder of the WEKA installation.

For learning purposes, select any data file from this folder. The contents of the file would be loaded in the WEKA environment. We will very soon learn how to inspect and process this loaded data. Before that, let us look at how to load the data file from the Web.

Developed by the University of Waikato, New Zealand, Weka stands for Waikato Environment for Knowledge Analysis.

Loading Data from Web

Once you click on the **Open URL ...** button, you can see a window as follows –

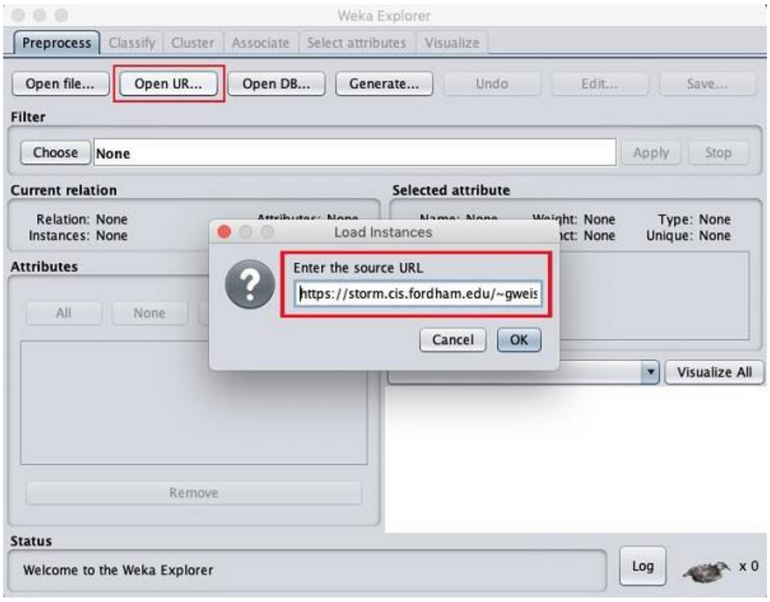

Figure 12:Loading Data from Web

We will open the file from a public URL Type the following URL in the popup box –

https://storm.cis.fordham.edu/~gweiss/data-mining/weka-data/weather.nominal.arff

You may specify any other URL where your data is stored. The **Explorer** will load the data from the remote site into its environment.

Loading Data from DB

Once you click on the **Open DB** ... button, you can see a window as follows −

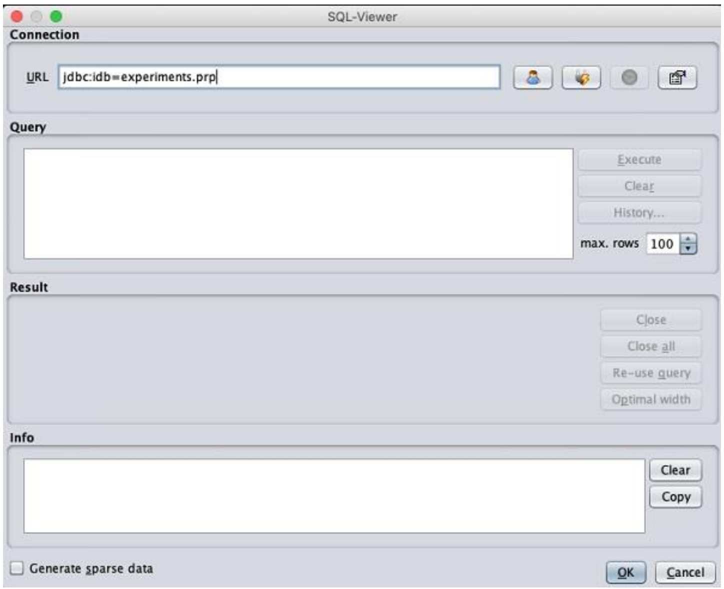

Figure 13: Loading of Data Using DB

Set the connection string to your database, set up the query for data selection, process the query, and load the selected records in WEKA.WEKA supports a large number of file formats for the data. The types of files that it supports are listed in the drop-down list box at the bottom of the screen. This is shown in the screenshot given below.

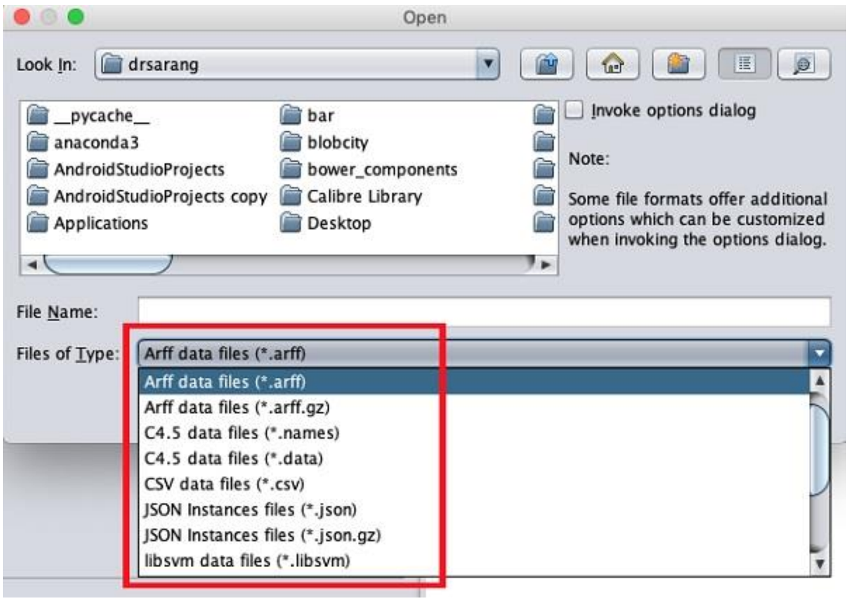

Figure 14: Files Supported by Weka

As you would notice it supports several formats including CSV and JSON. The default file type is Arff.

Arff Format

An **Arff** file contains two sections - header and data.

- The header describes the attribute types.
- The data section contains a comma-separated list of data.

As an example for Arff format, the **Weather** data file loaded from the WEKA sample databases is shown below −

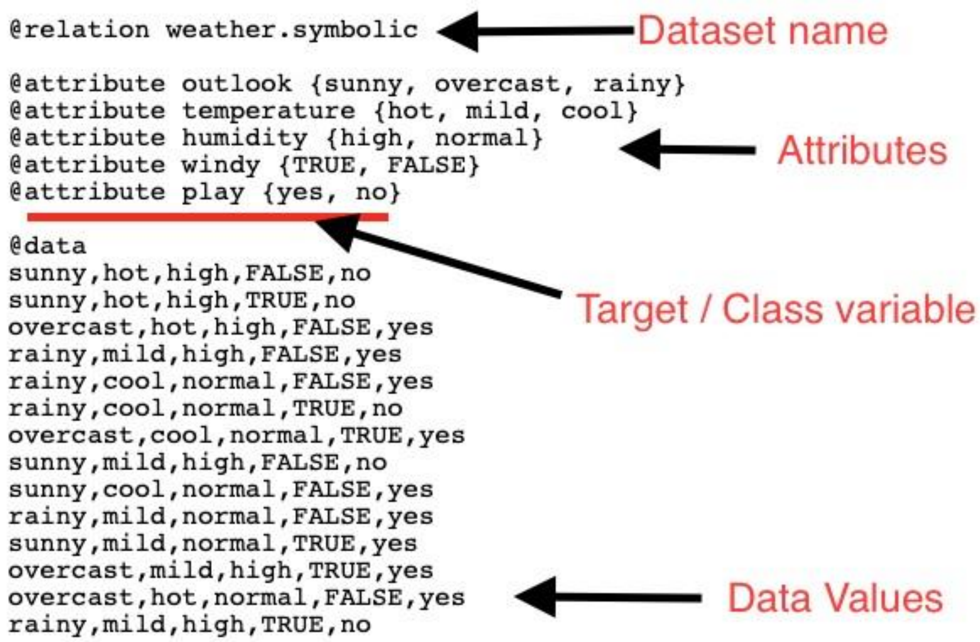

Figure 15:ARFF File

From the screenshot, you can infer the following points −

- The @relation tag defines the name of the database.
- The @attribute tag defines the attributes.
- The @data tag starts the list of data rows each containing the comma-separated fields.

The attributes can take nominal values as in the case of outlook shown here −

@attribute outlook (sunny, overcast, rainy)

The attributes can take real values as in this case −

@attribute temperature real

You can also set a Target or a Class variable called to play as shown here −

@attribute play (yes, no)

The Target assumes two nominal values yes or no.

Create your ARFF file and load it using Weka.

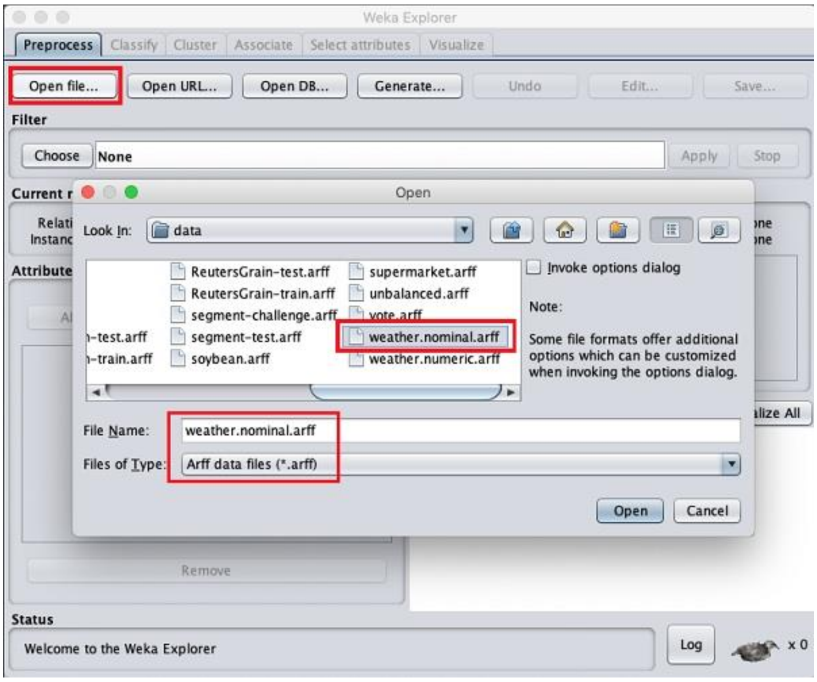

Figure 16:Opening a file using Weka

When you open the file, your screen looks like as shown here –

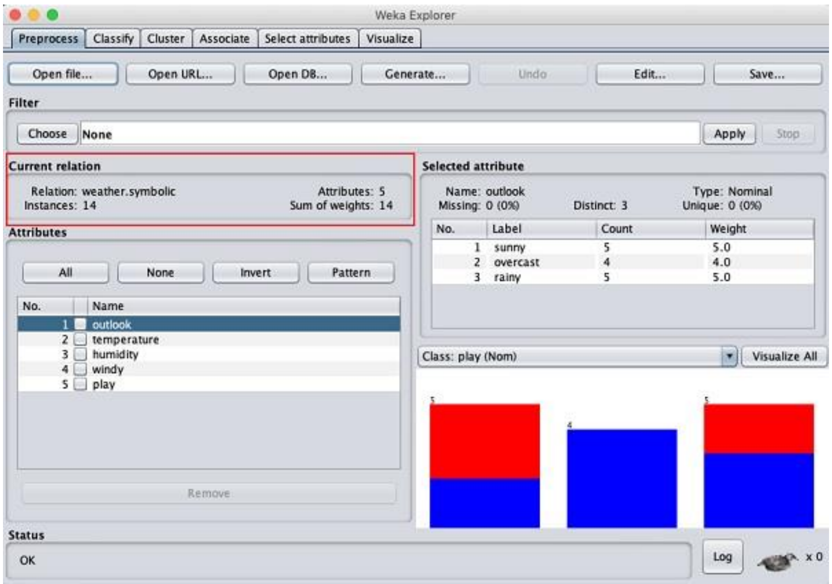

Figure 17: Pre-processing of Data

Understanding Data

Let us first look at the highlighted **Current relation** sub-window. It shows the name of the database that is currently loaded. You can infer two points from this sub window –

- There are 14 instances - the number of rows in the table.
- The table contains 5 attributes - the fields, which are discussed in the upcoming sections.

On the left side, notice the **Attributes** sub-window that displays the various fields in the database.

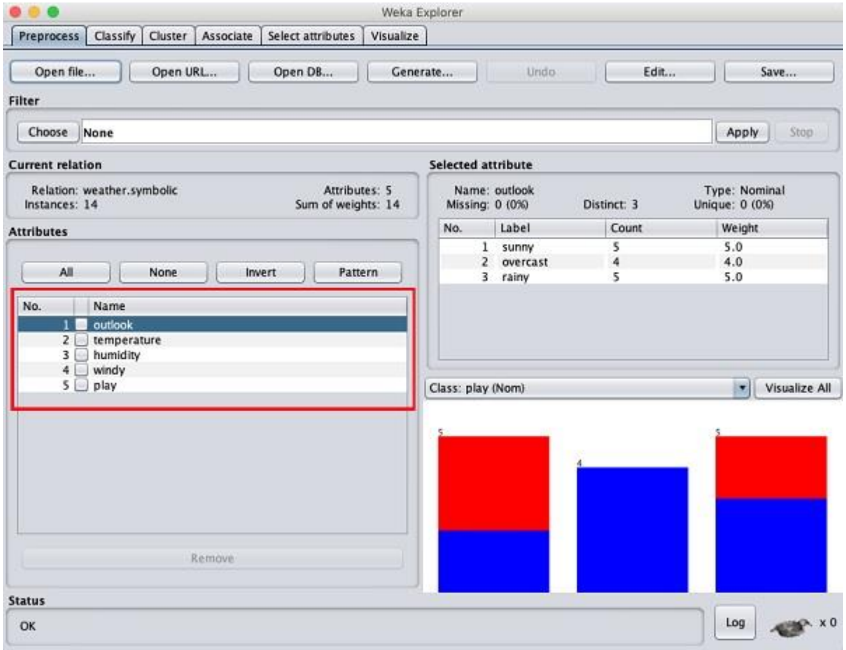

Figure 18: Selected Relation Attributes

The **weather** database contains five fields - outlook, temperature, humidity, windy, and play. When you select an attribute from this list by clicking on it, further details on the attribute itself are displayed on the right-hand side.

Let us select the temperature attribute first. When you click on it, you would see the following screen

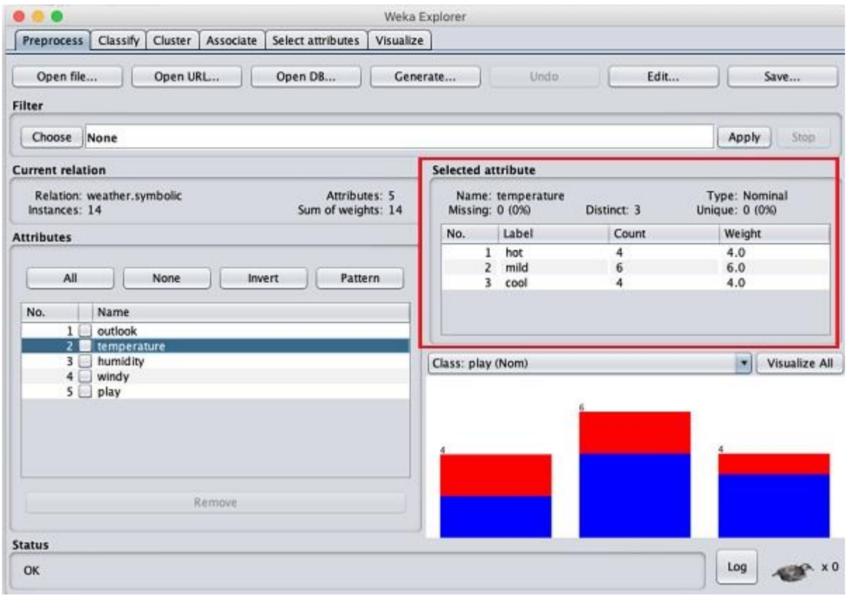

Figure 19: Statistics of Selected Attributes

In the **Selected Attribute** subwindow, you can observe the following −

- The name and the type of attribute are displayed.
- The type for the **temperature** attribute is **Nominal**.
- The number of **Missing** values is zero.
- There are three distinct values with no unique value.

- The table underneath this information shows the nominal values for this field as hot, mild, and cold.
- It also shows the count and weight in terms of a percentage for each nominal value.

At the bottom of the window, you see the visual representation of the **class** values.

If you click on the **Visualize All** button, you will be able to see all features in one single window as shown here −

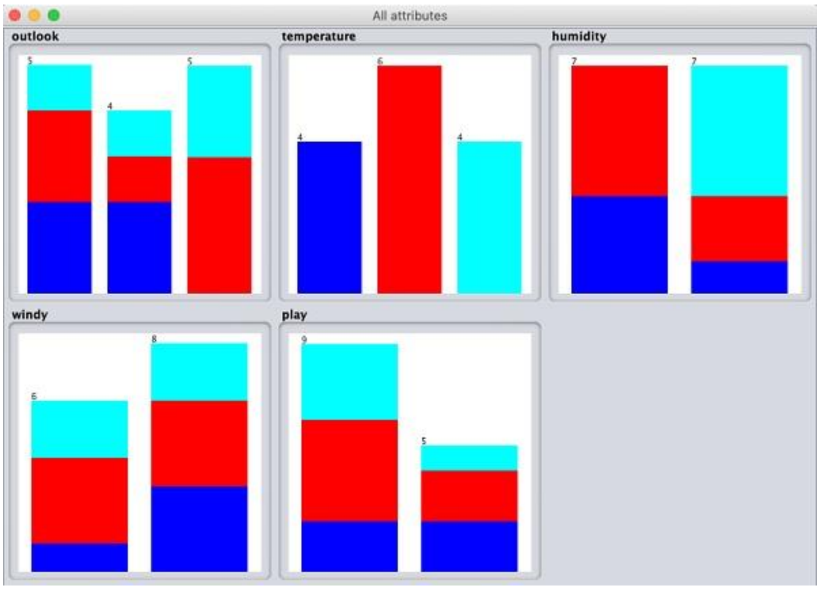

Figure 20: Visualization of Selected Data

Removing Attributes

Many a time, the data that you want to use for model building comes with many irrelevant fields. For example, the customer database may contain his mobile number which is relevant in analyzing his credit rating.

Figure 21:Attribute Removal

To remove Attribute/s select them and click on the **Remove** button at the bottom.

The selected attributes would be removed from the database. After you fully preprocess the data, you can save it for model building.

Next, you will learn to preprocess the data by applying filters to this data.

Applying Filters

Some of the machine learning techniques such as association rule mining requires categorical data. To illustrate the use of filters, we will use **weather-numeric.arff** database that contains two **numeric** attributes - **temperature** and **humidity**.

We will convert these to **nominal** by applying a filter to our raw data. Click on the **Choose** button in the **Filter** subwindow and select the following filter −

weka→filters→supervised→attribute→Discretize

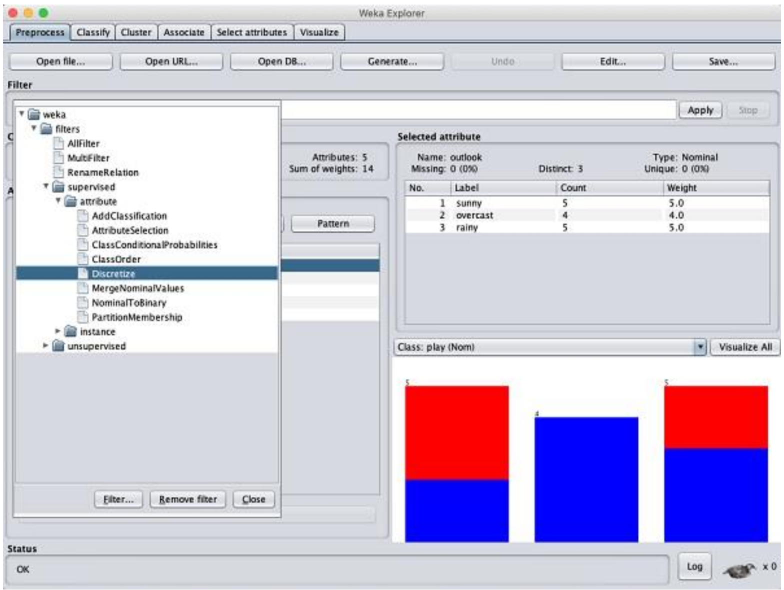

Figure 22:Application of Filters

After you are satisfied with the preprocessing of your data, save the data by clicking the **Save** ... button. You will use this saved file for model building.

5.6 Comparison between Rapid Miner and Weka

In the comparative study, I have concentrated on 2 of the commonly used tools:

- Rapid Miner
- Weka.

Features for Comparative Study

A comparative study is done based upon the following features:

- Usability
- Speed
- Visualization
- Algorithms supported
- Data Set Size
- Memory Usage
- Primary Usage
- Interface Type Supported

Table 3:Comparison

Features	Rapid Miner	Weka
Usability	Easy to use	Easiest to use.
Speed	Require more memory to operate.	Works faster on any machine.
Visualization	More options	Fewer options.
Algorithms supported	Classification and clustering	Classification and clustering
Data Set Size	Support small and large data sets.	Supports only small data sets.
Memory Usage	Requires more memory	Requires less memory and hence works faster.
Primary Usage	Data mining, predictive analysis	Machine learning
Interface type supported	GUI	GUI/CLI

Summary

- A perspective consists of a freely configurable selection of individual user interface elements, the so-called views.
- RapidMiner will eventually also ask you automatically if switching to another perspective.
- All work steps or building blocks for different data transformation or analysis tasks are called operators in RapidMiner. Those operators are presented in groups in the Operator View on the left side.
- One of the first steps in a process for data analysis is usually to load some data into the system. RapidMiner supports multiple methods for accessing datasets.
- It is always recommended to use the repository whenever this is possible instead of files.
- Open Recent Process opens the process which is selected in the list below the actions. Alternatively, you can open this process by double-clicking inside the list.
- WEKA supports many different standard data mining tasks such as data pre-processing, classification, clustering, regression, visualization and feature selection.
- The WEKA application allows novice users a tool to identify hidden information from database and file systems with simple to use options and visual interfaces.

Keywords

Process: A connected set of Operators that help you to transform and analyze your data.

Port: To build a process, you must connect the output from each Operator to the input of the next via a *port*.

Repository: your central data storage entity. It holds connections, data, processes, and results, either locally or remotely.

Operators: The elements of a Process, each Operator takes input and creates output, depending on the choice of parameters.

Filters. Processes that transform instances and sets of instances are called filters.

New Process: Starts a new analysis process.

Self Assessment Questions

1) __________________ is the central RapidMiner perspective where all analysis processes are created and managed.
a) Design Perspective
b) Result Perspective
c) Welcome Perspective
d) All of the Above
2) ____________________Opens the repository browser and allows you to select a process to be opened within the process Design Perspective.
a) Open Template
b) Online Tutorial
c) Open Process
d) Open Recent Process
3) Which of the following contains a large number of operators for writing data and objects into external formats such as files, databases, etc.
   a) Data Transformation
   b) Export
   c) Evaluation
   d) Import
4) Rapidminer comes with :
   a) Over 125 mining algorithms
   b) Over 100 data cleaning and preparation functions.
   c) Over 30 charts for data visualization
   d) All of the above
5) ___________________s designed to make the preparation of data much easier
   a) RapidMiner Auto Model
   b) RapidMiner Turbo Prep
   c) RapidMiner Studio
   d) None
6) Which of the following kinds of problems can be resolved with Auto Model.
a) Prediction.
b) Clustering.
c) outliers.
d) All of the above
7) The ________________ allows the automatic selection of features to create a reduced dataset.
a) Attributes Selection
b) Cluster, or Associate.
c) Classification
d) None
8) Which of the following is/are the features of RapidMiner.
a) Application & Interface
b) Data Access
c) Data Exploration
d) All
9) Which of the following is/are the features of RapidMiner.
   a) attribute selection

b) Experiments

c) workflow and visualization.

d) All

10) Which of the following is the data mining tool?

a) RapidMiner

b) Weka.

c) Both a and b.

d) None

11) Two fundamental goals of Data Mining are ________.

a) Analysis and Description

b) Data cleaning and organizing the data

c) Prediction and Description

d) Data cleaning and organizing the data

12) What is WEKA?

a) Waikato Environment for Knowledge Learning

b) Waikato Environmental for Knowledge Learning

c) Waikato Environment for Knowledge Learn

d) None.

13) Which of the following statements about the query tools is correct?

a) Tools developed to query the database

b) Attributes of a database table that can take only numerical values

c) Both and B

d) None of the above

14) Which one of the following refers to the binary attribute?

a) The natural environment of a certain species

b) Systems that can be used without knowledge of internal operations

c) This takes only two values. In general, these values will be 0 and 1, and they can be coded as one bit

d) All of the above

15) Data mining tools that exist in the market are

a) Weka.

b) Rapid Miner.

c) Orange

d) All of the above

Answers: Self Assessment

1.	A	2.	C	3.	B	4.	D	5.	B
6.	D	7.	A	8.	D	9.	D	10.	C
11.	C	12.	A	13.	A	14.	C	15.	D

Review Questions

Q1) Explain different perspectives available in RapidMiner Studio?

Q2) Differentiate between RapidMiner and Weka by considering different features.

Q3) Explain the various functions available under explorer in Weka?

Q4) Elaborate on the RapidMiner GUI in detail?

Q5) Write down the different methods of creating a repository in rapidMiner?

Further Readings

Witten, I. H., Frank, E., Trigg, L. E., Hall, M. A., Holmes, G., & Cunningham, S. J. (1999). Weka: Practical machine learning tools and techniques with Java implementations.

Markov, Z., & Russell, I. (2006). An introduction to the WEKA data mining system. ACM SIGCSE Bulletin, 38(3), 367-368.

Kotu, V., & Deshpande, B. (2014). Predictive analytics and data mining: concepts and practice with rapidminer. Morgan Kaufmann.

Hofmann, M., &Klinkenberg, R. (Eds.). (2016). RapidMiner: Data mining use cases and business analytics applications. CRC Press.

Web Links

https://www.cs.waikato.ac.nz/ml/weka/

https://www.analyticsvidhya.com/learning-paths-data-science-business-analytics-business-intelligence-big-data/weka-gui-learn-machine-learning/

https://storm.cis.fordham.edu/~gweiss/data-mining/weka.html

https://docs.rapidminer.com/9.5/server/configure/connections/creating-other-conns.html

https://docs.rapidminer.com/latest/studio/connect/